

Traverse Adjustment in Trimble Business Center

A step-by-step guide to importing conventional total-station data, defining a traverse, reviewing misclosures, and adjusting it in TBC with the compass, transit, or least-squares method.

Level Field and office survey staff

Software Trimble Business Center

Time about 30 minutes

01 • OVERVIEW

Before you start

A traverse adjustment spreads the measurement error found at closure across the traverse legs, so the final coordinates are consistent with fixed control. In TBC a single command, **Adjust Traverse**, both creates the traverse and adjusts it with the Compass (Bowditch) or Transit rule. ^[1] ^[4] TBC can also run a full **least-squares network adjustment**. ^[9]

Closed loop

Starts and ends on the same known point, with a known backsight azimuth. Gives a check on angles and distances, but cannot detect a scale or azimuth error in the starting control.

Closed connecting

Starts on one known pair of points and ends on a different known pair. Preferred, because it checks against independent control.

Open traverse

Does not close on known control. You can't compute a misclosure, so there is nothing to adjust. Avoid open traverses for control work.

You will need

- Trimble Access job files (.job or JobXML .jxl) or other supported raw data from the data collector.
- Published coordinates for at least the start point and backsight, and for the closing point and foresight on a connecting traverse.
- The project's datum, coordinate system, and units, plus the accuracy standard the job must meet (contract, agency such as WSDOT, or ALTA/NSPS).

- A TBC installation appropriately licensed for the survey and adjustment commands used here. Trimble's help doesn't tie these commands to a specific license tier, so check with your CAD/software administrator.

VERSION NOTE

Menu locations below follow the ribbon layout in current TBC releases. If your version looks different, search TBC for the command name, for example "Adjust Traverse."

Superscript numbers such as ^[1] link to the [References](#). Steps with a Trimble citation were checked against Trimble's online TBC help. Steps without one are standard office practice, so check them in your copy of TBC.

- † A menu path, button, field, or option that is *not* confirmed by the cited TBC help. It's shown as a typical location; verify it in your TBC version.
- ‡ Taken from Trimble Access or FieldLink help. These are *reference only* and may not be the exact TBC field name or option.

■ 02 · PROJECT

Project setup

1 Create a new project

Choose **File > New Project** † and pick the Parametrix survey template if one exists for your office. Otherwise use the default metric or US survey foot template.

2 Set the coordinate system

Go to **Project Settings > Coordinate System** †. Select the zone, for example *NAD83(2011) / Washington North (US ft)*, and the geoid model the project calls for.

3 Check units and the grid/ground setting

In **Project Settings > Units** †, confirm distance (US survey ft or m), angle (DMS), and display precision. If the project is on ground coordinates, confirm the combined scale factor or project-to-ground settings before importing anything.

4 Set computation options

Under **Project Settings > Computations**, confirm whether distances are reduced to grid or ellipsoid, and whether curvature and refraction corrections are applied. † The Computations settings page itself is documented in Trimble's help. ^[5]

5

Set the default standard errors

Open the **Default Standard Errors** section of Project Settings and check the values for both total station and GNSS observations before importing. ^[16] These are the a priori errors that weight every observation in a network adjustment, so they directly affect your results. For total station data, TBC takes its starting values from the imported job file. ^[16] Never leave GNSS (or any) standard errors at 0.00; that causes major problems in the adjustment. See [Least-squares alternative](#) for recommended values.

6

Save the project

Save it to the project folder using Parametrix's standard naming convention before you import data.

WATCH OUT

The wrong vertical units or grid/ground setting is the most common cause of a traverse that “closes badly” when the field work was actually fine. Check these settings before you import.

03 · DATA

Import field data

1

Open the Import pane

Home > Import [†]. You can also drag and drop .jxL files straight into the open project. [†]

2

Browse to the data folder

Select the .job or .jxL files and click **Import** [†]. If TBC warns about a coordinate system mismatch, click **Details** [†] to see a side-by-side comparison of the project and job coordinate systems. Keep the *project* coordinate system unless you know the field job is correct.

3

Check the import

In the **Project Explorer**, expand *Station Setups* [†] and *Points*. Each traverse station should have one setup (or more if you observed it in multiple sessions), with the correct backsight.

4

Check the shape in Plan View

This is a quick visual check for blunders before you go any further. Look for:

- **A station or line that sticks out** from where it should be. This usually means a wrong point ID, the wrong backsight, or a typo in a height or distance.
- **A station plotted at or near 0,0**. TBC had no starting coordinates for it, so it couldn't be computed.
- **A loop that doesn't come back to its start**. On a closed loop, the last leg should end very close to the starting point.

Set control points

The adjustment holds your control fixed and moves everything else. TBC needs to know which coordinates you want to hold.

1 Enter or import control coordinates

Use the **Create Point[†]** command, or import a control CSV.

2 Set the coordinate quality to Control

Select the point in the Project Explorer, open its **Properties** pane, and set the coordinate quality to **Control** (horizontal, and vertical if you are holding elevation). The quality levels, from highest to lowest, are *Control*, *Survey*, *Mapping*, and *Unknown*.^[6] ^[7] To change several points at once, Ctrl-select them in the Project Explorer and change the quality in Properties.^[8]

3 Match the point IDs

Control IDs must match exactly the IDs used in the field (for example CP1, not CP-1). If they don't, rename the points or merge the duplicates.

4 Verify the backsight azimuth

Compute an inverse between the start point and the backsight (the **Inverse[†]** command) and compare it to the published azimuth or bearing.

FIXED VS. ADJUSTED END STATIONS

TBC holds a start or end station **fixed** when its position and orientation are known without the rest of the traverse, such as a setup over a known point or a resection. A start or end station positioned only from the traverse itself is **adjusted**. If both ends are that type, only the backsight and foresight are known, and Trimble calls this the weakest kind of traverse.^[2]

Review raw observations

Fix blunders *before* you adjust. An adjustment spreads error around; it can't fix a bad observation.

CHECK	WHERE IN TBC [†]	WHAT TO LOOK FOR
Instrument & target heights	Station Setup properties; observation properties	Missing or 0.000 HI/HT, or a rod height that doesn't match the field notes
Prism constant	Observation properties › Target	Mixed constants (for example 0 mm vs –30 mm) between setups
Rounds / direct & reverse	Station setup observations in the Project Explorer	Large residuals in the mean turned angle; rounds outside spec
Backsight check	Station Setup › Backsight observation	Backsight distance or elevation residual well beyond the instrument's accuracy
Duplicate observations	Project Explorer › Points	The same point shot from both directions; reciprocal distances should agree within a few hundredths
Atmospheric / PPM	Station Setup properties	Temperature and pressure entered (or 0 ppm) applied the same way at every setup

To disable a bad observation, select it and disable it in its properties[†]. Don't delete it, because you want an audit trail.

Create the traverse

TBC has no separate “define” step. You create, edit, and adjust a traverse in the same **Adjust Traverse** command pane.^[1] If the traverse was already adjusted in Trimble Access, it comes in with the job, and you can use the same command to edit the station list or settings and re-adjust it.^[1]

1 Check the traverse project settings

Open **Project Settings › Computations › Traverse**.^[1] Here you choose whether traverses are adjusted automatically or manually. In manual mode, TBC re-checks the traverse on every project compute but doesn't change the adjusted coordinates. It flags the traverse for re-adjustment when the change exceeds the horizontal and vertical tolerances you set.^[5]

2 Open Adjust Traverse

Survey › Optical › Adjust Traverse.^[1]

3

Name and create the traverse

In **Traverse name**, enter a name that includes the job and date, for example PX-2601_Loop1_2026-10-08, then click **Create**. ^[1]

4

Select stations in order

Add the station setups in the order they were occupied: the start station, the intermediate stations, then the end station. Each intermediate station must have a foresight to the next station. ^[2]

5

Set the adjustment options

Choose the angular, distance, and elevation methods described in the [next section](#).

6

Preview Results

Click **Preview Results** to see the closure before anything is written to the project. ^[1] The Traverse Adjustment Report's **Before Adjustment** section shows the same misclosures, precision values, and traverse length. ^[3]

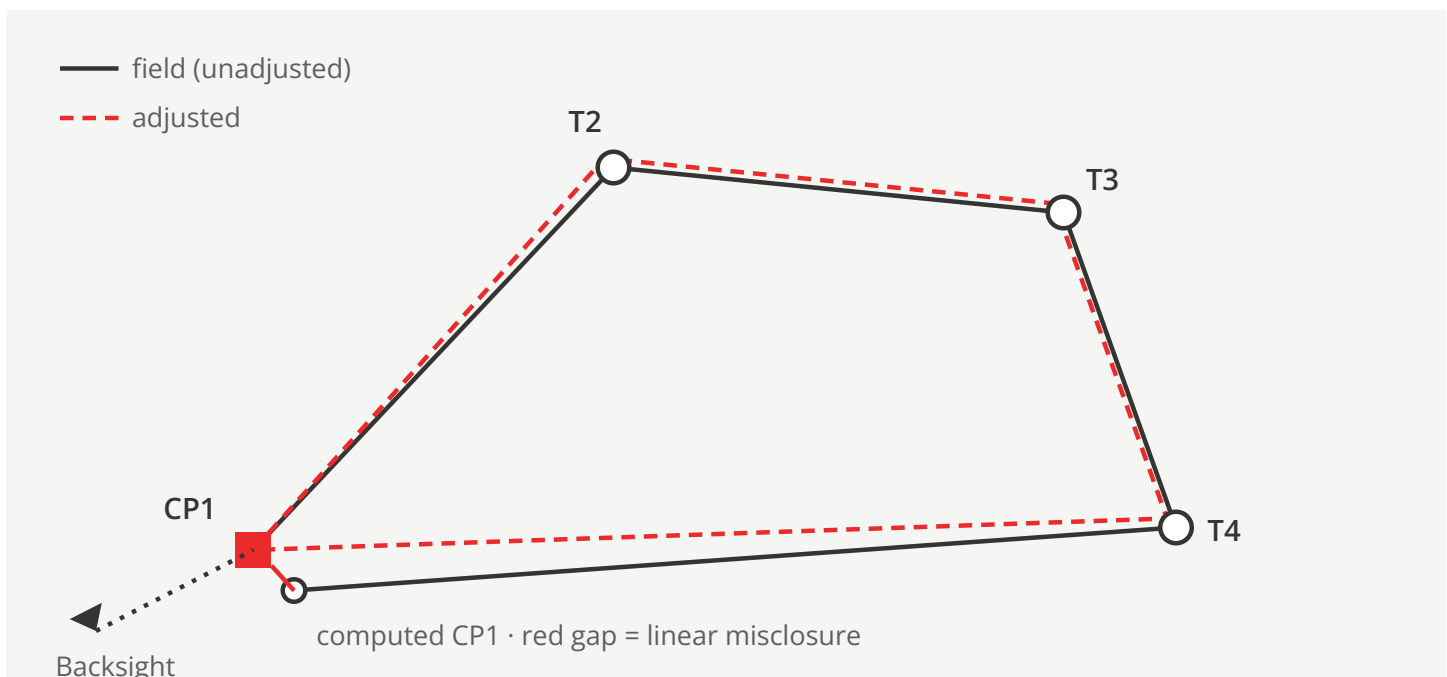


Figure 1: In a closed loop, the computed coordinates of the closing point miss the true control position. That vector is the linear misclosure. (Not to scale; error exaggerated.)

STOP AND CHECK

Compare the angular misclosure to the allowable value, usually $K \cdot \sqrt{n}$ seconds, where n is the number of angles and K comes from your specification. ^{[11] [13] [14]} If it's over the allowable value, or the precision ratio is below spec, go back to [Review raw observations](#). Don't adjust a traverse that fails closure.

Adjust & review

TBC applies the angular correction first and then distributes the linear misclosure. ^[4] ^[10]

1 Choose the angular distribution

Choose how TBC spreads the angular misclosure across the stations: in proportion to the leg distances, or in equal amounts at each station, which works best when the legs are similar in length. [‡] ^[10] (‡ These option names come from Trimble Access and FieldLink. The labels in the TBC pane may differ.)

2 Choose the linear (distance) adjustment

TBC's help documents two methods: ^[4]

- **Compass (Bowditch):** adjusts latitudes and departures in proportion to the length of each leg. ^[4]
This is the default choice when angles and distances are measured with similar precision, which is true of modern total stations.
- **Transit:** Trimble's definition is that it adjusts latitudes and departures in equal proportions or in proportion to the lengths of the lines. ^[4] Textbooks reserve it for cases where angles are much more precise than distances. ^[11] ^[15]

3 Choose the elevation adjustment

Distribute the vertical misclosure in proportion to distance or in equal proportions [‡] ^[10], or leave elevations unadjusted if you are carrying elevations from a separate level loop.

4 Apply

When the preview is within spec, click **Apply**. ^[1] TBC writes adjusted coordinates to the traverse stations. Fixed start and end stations are not changed. ^[2]

5 Open the Traverse Adjustment Report

Click the **Traverse Adjustment Report** icon on the Adjust Traverse pane toolbar, or open it from the Reports list in the Quick Access Toolbar. ^[3] Review these sections:

- **Before Adjustment:** raw misclosures, precision, and length
- **After Angular Adjustment** and **After Distance Adjustment:** what's left after each correction
- **End-Point Orientations** and **Fixed points:** confirm TBC held the control you expected
- **Station Adjustments:** the correction applied to each station, with a plan diagram ^[3]

ADJUST BEFORE YOU BUILD

Topo shots are computed from their station's coordinates, so adjust the traverse *before* you build surfaces or linework, and check a few sideshots afterward. Also note that merging a point in a manually adjusted traverse removes all of its traverse-adjusted coordinates and its report. ^[5]

08 • THE MATH

Worked example: compass rule

This is a four-sided closed loop starting and ending at **CP1** (N 5000.000, E 5000.000), in US survey feet. The azimuths shown have already been angularly balanced.

LEG	AZIMUTH	DIST (FT)	LATITUDE	DEPARTURE
CP1 → T2	45°00'00"	412.38	+291.597	+291.597
T2 → T3	135°00'04"	385.09	-272.305	+272.294
T3 → T4	224°59'57"	412.33	-291.566	-291.557
T4 → CP1	315°00'05"	385.16	+272.356	-272.343
Σ		1594.96	+0.082	-0.009

$$\text{Linear misclosure} = \sqrt{(0.082^2 + 0.009^2)} = 0.082 \text{ ft}$$

$$\text{Precision ratio} = 1594.96 / 0.082 \approx 1 : 19,400$$

$$\text{Compass correction (leg } i\text{): } C_{\text{lat}} = -\Sigma\text{Lat} \times (D_i / \Sigma D) \quad \cdot \quad C_{\text{dep}} = -\Sigma\text{Dep} \times (D_i / \Sigma D)$$

STATION	C _{LAT}	C _{DEP}	ADJUSTED N	ADJUSTED E
T2	-0.021	+0.002	5291.576	5291.599
T3	-0.020	+0.002	5019.251	5563.895
T4	-0.021	+0.002	4727.664	5272.341
CP1 (close)	-0.020	+0.002	5000.000	5000.000

The adjusted traverse closes exactly on CP1. A compass adjustment of the same data in TBC should produce the same values; compare them against the report's **Station Adjustments** table. ^[3] The method follows standard textbook practice. ^{[11] [15]} Working one traverse by hand like this is a good way to confirm you understand what the software is doing.

Least-squares alternative

Use the **Adjust Network** command ^[9] instead of (not in addition to) a compass adjustment when the traverse has redundant measurements, such as cross-ties, reciprocal shots, multiple loops, or GNSS-tied control.

1 Set the standard errors

Confirm the a priori errors in the **Default Standard Errors** section of Project Settings (see [Project setup](#)). For total station data, TBC starts with the values from the imported .job/.jxl file. ^[16] Enter values that match your equipment, for example 1" or 2" for angles and 1 mm + 1.5 ppm for EDM. Set the GNSS standard errors too; values left at 0.00 cause major problems.

PARAMETRIX PRACTICE

Use **0.01 ft** for centering and **0.01 ft** for the height measurement (measure-up), for both total station and GNSS. The exact field names in TBC aren't confirmed here. [†]

2 Select the fixed control

On the Adjust Network pane tab that lists the control-quality coordinates [†], check **2D**, **h** (ellipsoid height), or **e** (elevation) for each control coordinate you want to hold. ^[9]

3 Run a minimally constrained adjustment first

Trimble's adjustment workflow starts with a **minimally constrained** adjustment, before any additional control is held. ^[9] Fix only the minimum needed to position and orient the network. For a total-station-only network, that is usually one point and one azimuth. ^[12] This tests the observations on their own, so a bad control value can't hide in, or distort, the results.

4 Evaluate the Reference Factor and Chi Square test

The Chi Square test checks whether your a priori error estimates agree with the adjustment. ^[9] If it fails, look for blunders and systematic errors first. ^{[9] [12]} A failed test is not, on its own, proof of a bad adjustment, or a reason to change the weighting. Adjustments sometimes "fail" while the residuals, control comparisons, and independent checks all show the results are sound. Change the weighting only when there's a significant reason to, and document why. When you do, the **Weighting** tab is used to improve the a priori error estimates, and Trimble's goal when adjusting the weighting scalars is to bring the **Reference Factor** toward 1.00. ^[9] Re-adjust and re-evaluate both values together. Don't just rescale errors until the test passes; a scalar far from 1 is telling you something about the data or the error model. Flag any residuals over about 3σ . ^[12] See [Read the Network Adjustment Report](#).

5 Add the additional control constraints

Once the minimally constrained adjustment is clean, hold the rest of the trusted control and re-adjust. Large shifts at this stage point to a problem with the control, not the observations.

DON'T MIX METHODS

Pick one method per traverse. A network adjustment saves new coordinate records with quality *Adjusted* or *Fixed in network adjustment* and makes them each point's final value. ^[9]

Survey > Network > Clear Adjustment Results ^[9] does more than remove the adjusted coordinates. It also clears the adjustment transformation parameters, the coordinate fixes and adjustment flags, resets the weighting scalars to 1.00, removes the error ellipses, and recomputes the project. Running both methods on the same stations leaves two sets of results, so decide which one is the record.

■ 10 • ANALYSIS

Read the Network Adjustment Report

The report isn't just a pass/fail check. Trimble describes it as the tool for deciding which observations to disable, which control to fix, and which settings to change before you re-adjust. ^[17] Open it with the **Network Adjustment Report** button on the Adjust Network pane toolbar. It opens in your web browser. ^[9]

REPORT SECTION	WHAT IT TELLS YOU	WHAT TO LOOK FOR
Adjustment Statistics	Number of iterations, whether the adjustment passed or failed and why, the Reference Factor, the Chi Square test (95%), and degrees of freedom ^[17] ^[18]	A Reference Factor near 1.00. Degrees of freedom above zero; with zero there's no redundancy, so nothing is being checked. ^[18]
Adjusted Observations (GNSS, total station, leveling)	Each observation's standard error, residual (how much it had to be adjusted), and standardized residual, sorted worst first ^[17]	Observations in red failed the Tau test and are outliers. Trimble says to examine each one and justify keeping it. ^[17] ^[18]
Control Coordinate Comparisons	The difference between each control point's known coordinates and its adjusted coordinates ^[17]	One control point with a much larger difference than the rest points to a bad control value or the wrong point. Compare the minimally and fully constrained runs.
Adjusted Geodetic Coordinates	Final adjusted positions. The Fixed column shows which coordinates were held. ^[17]	Only the control you intended to hold should be fixed.
Error ellipses [†]	The size and orientation of each point's positional uncertainty ^[17]	Large or very elongated ellipses mean weak geometry, such as a point tied in by only one observation.

Working through the report

1

Start with Adjustment Statistics

Note the Reference Factor, the Chi Square result, and the degrees of freedom before you look at anything else.

2

Work the outliers from the top down

Start at the top of the Adjusted Observations lists, where the worst standardized residuals are. ^[17] For each red observation, look for a cause: a wrong instrument or target height, a wrong point ID, a bad backsight, or a misidentified control point. Fix the cause where you can. Disable an observation only when you can justify it, not just because it's red.

3

Re-adjust and compare

After each change, re-run the adjustment and check that the statistics and control comparisons improved for the reason you expected. Change one thing at a time.

4

Decide and document

When the residuals are reasonable and the control comparisons agree, accept the adjustment and save the report with the survey QA file. Record any disabled observations and any weighting changes, and why.

PARAMETRIX PRACTICE

A “failed” Chi Square test doesn't automatically mean the adjustment is wrong. Experienced reviewers sometimes accept adjustments that fail the test when the residuals are small, there are no unexplained outliers, and the results agree with independent checks. Don't change the weighting unless there's a significant reason to, and note the decision in the survey narrative.

■ 11 • DELIVERABLES

Reports & QA

- **Traverse Adjustment Report:** misclosures, corrections, and adjusted coordinates. ^[3] Attach it to the survey QA file.
- **Network Adjustment Report:** open it from the Adjust Network pane toolbar. It includes residuals, error ellipses, and statistical tests (least squares only). ^[9]
- **Point Derivation Report**[†]: confirms which coordinate record each point is using.
- Export the adjusted coordinates with **Home > Export** [†] using the office's standard PNEZD CSV format, and note the adjustment method and precision ratio in the survey narrative.

Troubleshooting

SYMPTOM	LIKELY CAUSE	FIX
Angular misclosure is huge (degrees)	Wrong backsight point, or an orientation error at setup	Check the backsight ID and azimuth on each Station Setup
Misclosure is roughly one leg length	A station was skipped or selected out of order in Adjust Traverse	Edit the station list in Adjust Traverse so the stations are in occupation order ^[1]
Error grows with distance (scale-like)	Grid/ground mix-up, wrong combined scale factor, or ppm not applied	Recheck the Project Settings computation and scale settings, and the setup atmospherics
Elevation misclosure only	Wrong HI/HT, or a prism height changed without being recorded	Compare to field notes and edit the heights in properties
Every station gets adjusted, or closure looks meaningless	Neither end station is fixed. They aren't over Control-quality points with a known backsight or foresight ^[2]	Set the control to Control quality and confirm the end-point orientations in the report ^[3]
Traverse adjustment disappeared	A point in a manually adjusted traverse was merged ^[5]	Re-run Adjust Traverse and regenerate the report
Sideshots didn't move	The shots came from a different setup of the same point	Merge the setups, or include that setup in the traverse

Checklist

- ☐ Coordinate system, geoid, units, and grid/ground confirmed
- ☐ Default standard errors set for total station and GNSS (none left at 0.00)
- ☐ Field data imported; plan view looks right
- ☐ Control points set to Control quality, with IDs matching
- ☐ HI/HT, prism constants, and atmospherics checked
- ☐ Traverse created; angular and linear closure within spec

- ☐ Adjustment run (one method only); report reviewed, saved, and any disabled observations or weighting changes documented
- ☐ Adjusted coordinates exported and adjustment noted in the narrative
- ☐ Reviewed by PLS / survey lead

■ 14 • SOURCES

References

Trimble help pages were checked in October 2026. Trimble updates its help with each TBC release, so confirm them against the version you have installed.

TRIMBLE BUSINESS CENTER HELP

- [1] Trimble. "Create, Edit, and Adjust a Traverse." *Trimble Business Center Help*.
<https://help.fieldsystems.trimble.com/tbc/1927.htm>
- [2] Trimble. "Understanding Traverses." *Trimble Business Center Help*.
<https://help.fieldsystems.trimble.com/tbc/16947.htm>
- [3] Trimble. "Run a Traverse Adjustment Report." *Trimble Business Center Help*.
https://help.fieldsystems.trimble.com/tbc/16994_1.htm
- [4] Trimble. "traverse adjustment" (glossary). *Trimble Business Center Help*.
<https://help.fieldsystems.trimble.com/tbc/3013.htm>
- [5] Trimble. "Computation Settings." *Trimble Business Center Help*.
<https://help.fieldsystems.trimble.com/tbc/3341.htm>
- [6] Trimble. "How Coordinate Quality for a Point is Displayed." *Trimble Business Center Help*.
<https://help.fieldsystems.trimble.com/tbc/10262.htm>
- [7] Trimble. "Understanding Point Types." *Trimble Business Center Help*.
<https://help.fieldsystems.trimble.com/tbc/6159.htm>
- [8] Johnson, E. "TIP #148: How to Change the Quality of Multiple Points at Once." *Trimble Community*, 2021.
community.trimble.com
- [9] Trimble. "Adjust a Network" and "Workflow: Adjust the Network." *Trimble Business Center Help*.
<https://help.fieldsystems.trimble.com/tbc/396.htm> ·
<https://help.fieldsystems.trimble.com/tbc/12079.htm>

TRIMBLE FIELD SOFTWARE (§ REFERENCE ONLY, NOT TBC FIELD NAMES)

- [10] Trimble. "Traverse calculations." *Trimble Access Help*. help.fieldsystems.trimble.com/trimble-access/latest/en/cogo-traverse.htm. See also "Traverse (Optional Module)," *Trimble FieldLink Help*: help.fieldsystems.trimble.com/fieldlink/traverse.htm
-

TEXTBOOKS & STANDARDS

- [11] Ghilani, C. D., & Wolf, P. R. *Elementary Surveying: An Introduction to Geomatics*, 16th ed. Pearson, 2021. Chapter 10, "Traverse Computations" (angular misclosure, latitudes and departures, compass rule, relative precision).
-
- [12] Ghilani, C. D. *Adjustment Computations: Spatial Data Analysis*, 6th ed. Wiley, 2017. Least-squares traverse and network adjustment, reference variance, χ^2 test, and blunder detection.
-
- [13] Washington State Department of Transportation. *Highway Surveying Manual*, M 22-97 (current revision). Survey accuracy standards and closure requirements for WSDOT work.
-
- [14] NSPS / ALTA. *Minimum Standard Detail Requirements for ALTA/NSPS Land Title Surveys* (current edition). Relative positional precision requirements.
-
- [15] Mahun, J. "Adjustment Methods Comparison" (Compass, Transit, Crandall). *Open Access Surveying Library*. jerrymahun.com
-
- [16] Blecha, J. "Tip of the Week #29: Sources of Default Standard Errors." *Trimble Community*, 2020. community.trimble.com
-

TRIMBLE BUSINESS CENTER HELP (NETWORK ADJUSTMENT REPORT)

- [17] Trimble. "Run a Network Adjustment Report." *Trimble Business Center Help*. https://help.fieldsystems.trimble.com/tbc/398_1.htm
-
- [18] Trimble. "Network Adjustment Options." *Trimble Business Center Help*. <https://help.fieldsystems.trimble.com/tbc/3338.htm>
-

Parametrix

Internal training material · Traverse Adjustment in TBC · Verify menu paths against your installed TBC version